Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 13: You Already Proved These

The Algebra of Sets

Chartrand & Zhang, Section 2.2, Theorems 2.22 and 2.35

Name: _____ Date: _____

Where we are

Monday you saw that membership in a combined set is a logical statement about membership in the pieces. Today that observation does real work: **every law about sets is a law about statements, already proved in Week 1.** You will not learn a single new fact today. You will learn that you already own them.

By the end of today you should be able to

1. State the commutative, associative, and distributive laws for sets.
2. State De Morgan's Laws for sets.
3. Derive a set law from the corresponding law about statements.

1. The laws

Theorem 2.22

For all sets A, B, C:

Commutative:	$A \cap B = B \cap A$	$A \cup B = B \cup A$
Associative:	$(A \cap B) \cap C = A \cap (B \cap C)$	$(A \cup B) \cup C = A \cup (B \cup C)$
Distributive:	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Theorem 2.35 — De Morgan's Laws for sets

$$\overline{A \cup B} = \overline{A} \cap \overline{B} \qquad \overline{A \cap B} = \overline{A} \cup \overline{B}$$

Match each to the Week 1 result it came from.

Law about sets	Law about statements
$A \cap B = B \cap A$	$P \wedge Q \equiv Q \wedge P$ (Thm 1.22)
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	$P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$ (Thm 1.31)
$\overline{A \cup B} = \overline{A} \cap \overline{B}$	$\sim(P \vee Q) \equiv (\sim P) \wedge (\sim Q)$ (Thm 1.24)
$\overline{A \cap B} = \overline{A} \cup \overline{B}$	$\sim(P \wedge Q) \equiv (\sim P) \vee (\sim Q)$ (Thm 1.24)

Two of these you proved yourself

The $\wedge$-over-$\vee$ distributive law was Lecture 3's worked example. The $\vee$-over-$\wedge$ law was **Homework 2, Problem 3** — the one the arithmetic analogy predicted wrongly. Both now become set laws for free.

2. How the derivation actually goes

Here is De Morgan for sets, derived line by line from De Morgan for statements. This is the book's own argument.

$$\begin{aligned}
x \in \overline{A \cup B} &\equiv \sim\big(x \in A \cup B\big) && \text{definition of complement}\\
&\equiv \sim\big((x \in A) \vee (x \in B)\big) && \text{definition of union}\\
&\equiv \big(\sim(x \in A)\big) \wedge \big(\sim(x \in B)\big) && \text{De Morgan, Theorem 1.24}\\
&\equiv (x \in \overline{A}) \wedge (x \in \overline{B}) && \text{definition of complement}\\
&\equiv x \in \overline{A} \cap \overline{B} && \text{definition of intersection}
\end{aligned}$$

Since an element belongs to the left side exactly when it belongs to the right side, the two sets are equal.

The recipe

1. Translate "$x \in$ (set expression)" into a logical statement.
2. Apply a law you proved in Week 1.
3. Translate back into set notation.

Only step 2 does any mathematics. Steps 1 and 3 are dictionary lookups.

3. Your turn

Derive $\overline{A \cap B} = \overline{A} \cup \overline{B}$ the same way.

Step	Reason
$x \in \overline{A \cap B}$	(start)
$\equiv \sim\big((x \in A) \wedge (x \in B)\big)$	complement, then intersection
$\equiv \big(\sim(x \in A)\big) \vee \big(\sim(x \in B)\big)$	De Morgan (Thm 1.24)
$\equiv x \in \overline{A} \cup \overline{B}$	complement, then union

4. Checking a law with a membership table

An identity holds exactly when the two columns agree in every row. Verify $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

A	B	C	$B \cup C$	$A \cap (B \cup C)$	$A \cap B$	$(A \cap B) \cup (A \cap C)$
T	T	T	T	T	T	T
T	T	F	T	T	T	T
T	F	T	T	T	F	T
T	F	F	F	F	F	F
F	T	T	T	F	F	F
F	T	F	T	F	F	F
F	F	T	T	F	F	F
F	F	F	F	F	F	F

Columns five and seven agree in all 8 rows, so the identity holds.

This is the same table as Lecture 3

You built exactly these columns in Lecture 3 for $P \wedge (Q \vee R)$. Only the header row has changed. That is the entire content of today.

Exit ticket

1. Write $\overline{A \cup B}$ without an outer bar, and name the law.

2. Which Week 1 theorem gives $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$?

Before Friday

Read Section 2.2, Example 2.31 (pp. 65–66) — a set identity proved by showing each side is a subset of the other. That is Friday's method.

Self-check (not collected). Section 2.2, Exercises 9, 11, and 15 — odd-numbered, answers in the back.

Homework 5 is due Friday, September 18, 11:59 p.m.

Sections 3.1, 2.1, and 2.3. One PDF to Canvas with the collaboration and AI statement first.

Questions I still have